

# Yichen Hu

Email: ethhu@ucdavis.edu

Davis, CA, USA

## EDUCATION

---

- **University of California, Davis** Davis, CA, USA  
Ph.D. Candidate in Statistics Sep 2023 – Present  
Advisor: Dr. Xiukai Ding.  
Research interests: high-dimensional statistics, random matrix theory, manifold learning, graph networks, and applications to data science and machine learning.
- **University of California, Davis** Davis, CA, USA  
Master of Science in Statistics Sep 2021 – Jun 2023
- **Shanghai University of Finance and Economics** Shanghai, China  
Bachelor of Science in Statistics Sep 2016 – Jul 2020

## RESEARCH PROJECTS

---

Note: <sup>†</sup> denotes that authors are listed in alphabetical order.

- Xiukai Ding, **Yichen Hu**, Zhenggang Wang<sup>†</sup> (2025). Two Sample Test for Covariance Matrices in Ultra-High Dimension. *Journal of the American Statistical Association*, 120(552), 2210–2221.
- Xiukai Ding, **Yichen Hu**<sup>†</sup> (2025). On the Edge Eigenvalues of Sparse Random Geometric Graphs. Submitted to *Annals of Applied Probability*. Available at <https://arxiv.org/abs/2509.07372>.
- **Yichen Hu**, Xiukai Ding, Jian Kang (2026). Adaptive Sparsity-Aware Projection-Assisted Spectral Embedding for High-Dimensional Noisy Data with Nonlinear Structures. Submitted.
- Xiukai Ding, **Yichen Hu**, Jiahui Xie<sup>†</sup>. Bias-Corrected Multiplier Bootstrap Inference for Spectral Edges of Large Covariance Matrices. Submitted to *The Journal of the Royal Statistical Society, Series B*. Available at <https://arxiv.org/abs/2607.08089>.
- **Yichen Hu**, Qifeng Lin, Xiukai Ding, Rong Ma. Learning common structures from high-dimensional noisy biviews using cross-integral operator. In progress.

## AWARDS

---

- Peter Hall Graduate Student Research Award, Department of Statistics, University of California, Davis, 2026
- IMS Hannan Graduate Student Travel Award, 2026
- Teaching Recognition Award, Department of Statistics, University of California, Davis, 2025–26
- Teaching Recognition Award, Department of Statistics, University of California, Davis, 2024–25
- Teaching Recognition Award, Department of Statistics, University of California, Davis, 2023–24
- Outstanding Master's Student Award, Department of Statistics, University of California, Davis, 2021–22

## SOFTWARE

---

- **RMT4DS**: an R package for random matrix tools in data science.  
Available at CRAN: <https://cran.r-project.org/package=RMT4DS>
- **UHDtst**: an R package for two-sample covariance matrix testing in ultra-high-dimensional settings.  
Available at GitHub: <https://github.com/EthanHuStat/UHDtst>.

## TEACHING

---

- Teaching Assistant, Department of Statistics, University of California, Davis  
– STA 206: Statistical Methods for Research I

- STA 141B: Data & Web Technologies for Data Analysis
- STA 135: Multivariate Data Analysis
- STA 106: Applied Statistical Methods: Analysis of Variance
- STA 108: Applied Statistical Methods: Regression Analysis

## SERVICE

---

- Reviewer for *Electronic Journal of Statistics*.
- Reviewer for *Communications in Statistics – Theory and Methods*.